

Sarthak Singh

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Experience

Influenced Full Stack Developer — <i>Java, Spring Boot, React JS, Node.js, AWS, DynamoDB</i>	July 2024 – June 2025 Remote, USA
- Built an AI-powered marketplace connecting brands with inventory to micro-influencers, using a generative AI recommendation engine to rank and match influencers with brand campaigns using NLP-based profile analysis for collaborative filtering. - Built backend microservices with Spring Boot, implementing secure REST endpoints, custom JPA queries for database optimization, and dependency injection patterns for maintainable, and supporting 10K+ concurrent users with loosely-coupled architecture. - Implemented RBAC and AI-driven workflows for product listings and collaboration management, orchestrated deployment on AWS EKS with auto-scaling, configured load balancing and CloudWatch alerts, leveraging DynamoDB for high-performance data storage.	
IBM Software Development Engineer II — <i>Python, AWS, GO, Java, Docker, Kubernetes, PostgreSQL, Nginx</i>	July 2021 – Jan 2024 Kochi, India
- Led the migration of from ElasticSearch to OpenSearch, redesigning indexing strategies and query optimization layers using Python and FastAPI to expose search APIs, reducing operational costs by up to \$3M annually and improving search query performance. - Streamlined backend services in Golang for API management on IBM Cloud and AWS, leveraging Go's concurrency model with goroutines to handle high-throughput traffic routing via Nginx, reducing response times across distributed systems. - Led the migration of secrets from SOS Vault to IBM Secrets Manager, optimizing the secrets retrieval caching layer to reduce latency while pulling credentials, resulting in a more cost-effective secrets management infrastructure and earning me the IBM Cash Award. - Containerized microservices with Docker, orchestrated deployments on Kubernetes (EKS), and automated CI/CD pipelines with Jenkins, improving deployment efficiency and system observability using CloudWatch and Grafana.	
Informatica by Salesforce Software Engineer Intern — <i>Python, ETL, PostgreSQL, Tableau, SQLAlchemy, AWS S3</i>	Jan 2021 – July 2021 Bangalore, India
- Engineered Python-based ETL pipelines using Pandas and SQLAlchemy to automate data ingestion and transformation across multiple data sources, implementing retry logic and error handling to ensure pipeline reliability, reducing manual processing time by 40%. - Built an internal Python CLI tool using regex-based log parsing to triage ETL pipeline failures across multiple heterogeneous data sources, automatically generating structured error reports, cutting mean resolution time by 30%	

Project & Research

AI Powered Solutions	Oct 2025 – Dec 2025
- Voice Customer Support: Built a voice-enabled support assistant using GPT, HuggingFace Transformers, and ChromaDB, combining RAG with real-time speech interaction to deliver context-aware responses over a conversational voice interface. - Indian Law Assistant: Developed an AI-powered legal assistant leveraging RAG to ground LLM responses in Indian legal statutes and case law, integrating MCP for structured ingestion and processing of real-world legal documents to reduce hallucination. - GPT-OSS CritiqueLoop: Designed an automated critique-improve pipeline using Groq for ultra-fast inference on a 120B open-source model, iteratively refining LLM outputs through self-critique loops to improve response quality without human-in-the-loop intervention.	
Methane Optimizer Tool	Aug 2024 – Dec 2024
Led the web development of a ReactJS application to visualize the optimization of methane production in a Urea plant, visualizing methane output levels by tweaking control parameters to optimized values generated by underlying CNN deep learning models.	
Research Article (Journal of Environmental Management)	July 2022 – Jan 2024
Contributed to research on green procurement dynamics in the public sector, developing an assessment framework to help developing nations make data-driven decisions towards green environmental practices.	

Education

University of Maryland Master in Engineering, Software Engineering, Cumulative GPA: 3.6/4.0 Teaching Assistant <i>Introduction to Software Engineering ENPM611</i>	College Park, MD Jan 2024 – Dec 2025
Vellore Institute of Technology Bachelor of Technology in Computer Science and Engineering	Vellore, India June 2017 – May 2021

Technical Skills

Languages: Python, Go, Java, SQL, HTML, CSS, JavaScript, Bash
Tech Stack: FastAPI, Flask, Spring Boot, React, Redis, Kafka, Docker, Kubernetes, Jenkins, Terraform, Git, Grafana, Linux
Databases: PostgreSQL, MySQL, MongoDB, Cassandra, ChromaDB
Cloud: AWS (CloudWatch, EC2, EKS, IAM, S3, Lambda), Azure (Blob Storage, AKS, Azure Functions, AD, Monitor)
Concepts: Microservices, Distributed Systems, CI/CD, Object-Oriented Design, Unit Testing, Data Structures
AI: NLP (Transformer Models), LLM (Bert, Grok, Claude, GPT), Deep Learning & ML (PyTorch, Scikit-learn)